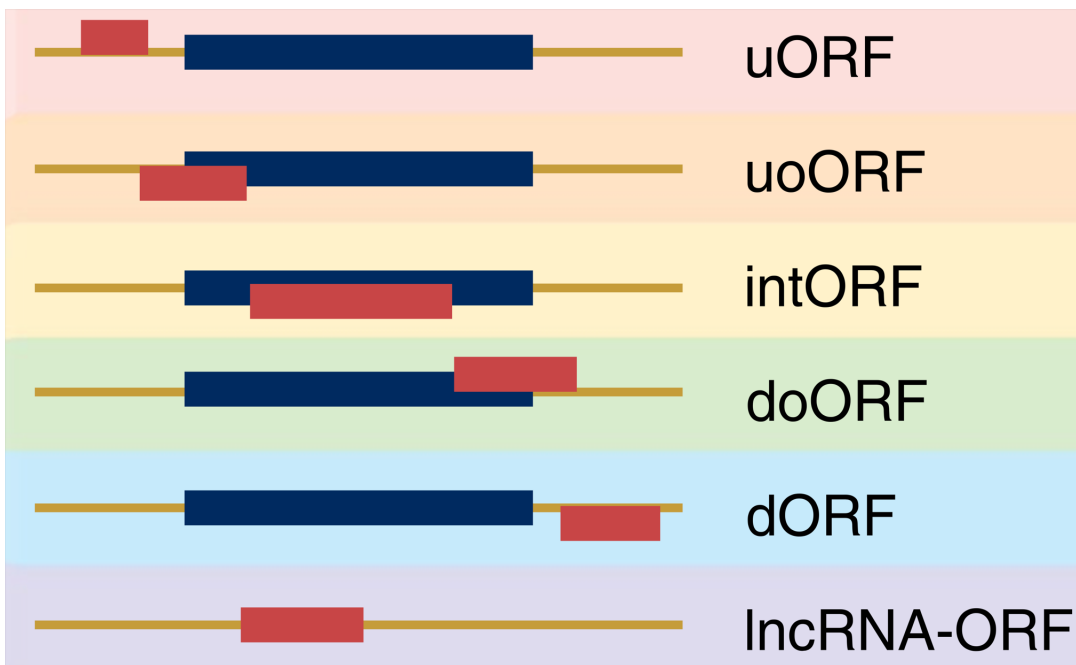


Non-canonical ORF Biotypes



Graphic from Mudge et al. *Nature Biotechnology* 2022

- uORF: Upstream ORF
 - Entirely in 5'-UTR
- uoORF: Upstream overlapping ORF
 - Starts in 5'-UTR; overlaps the CDS in alternate frame.
- intORF: Interior ORF
 - Entirely within CDS in alternate frame
- doORF: Downstream overlapping ORF
 - Starts overlapping the CDS in alternate frame; ends in 3'-UTR
- dORF: Downstream ORF
 - Entirely in 3'-UTR
- lncRNA-ORF: lncRNA ORF
 - ORF does not lie on or overlap an annotated mRNA
 - This biotype is used even if ORF is not on a lncRNA
- Mixed
 - None of the above conditions apply. For example, ORF overlaps two CDSs in different reading frames.